OpenCUA: Open Foundations for Computer-Use Agents

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Paper version: arXiv 2508.09123v3 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Critical details of the most capable computer-use agents remain closed.*

Open desktop data and training recipes are scarce; this paper releases data, code and models, and trains screenshot-based policies by supervised fine-tuning on human demonstrations.

Motivation: three gaps

1. **Closed recipes.** Top CUAs do not publish data or training.
2. **Narrow data.** Open GUI sets: mobile, web or grounding.
3. **State-action alone fails.** Plain SFT: 4.4% on OSWorld.

Evaluation setup

OSWorld-Verified · online

WindowsAgentArena · online

AgentNetBench · offline

Qwen2.5-VL 7B / 32B / 72B

15 / 50 / 100 steps

3 runs, temp 0

- **Online:** OSWorld-Verified, scored by the OSWorld team; **offline:** AgentNetBench, grounding sets.
- **Held out:** no overlap with OSWorld tasks; 72B adds 8k o3+Jedi rollouts.

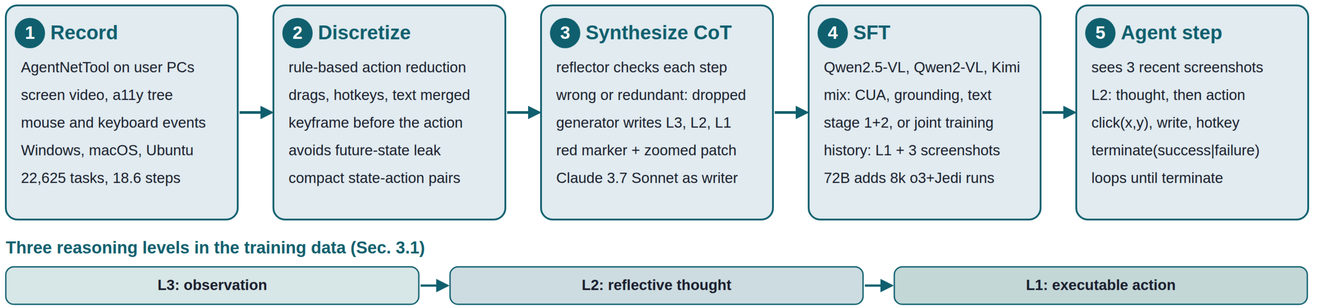
Our read (CUA Radar)

69	★★	online
score of 100	band	evidence level

- **Reports:** online eval; 3 runs; step budgets; baselines; cost; CoT ablation; code announced.
- **Flags:** held-out partial; reflector unvalidated.
- Pattern C21. Authors' numbers, not rerun.

Method: human demos, reflective long CoT, SFT

- **Sees / acts:** screenshots (3 as history) in; pyautogui-style click, write, hotkey, terminate out.
- **Data:** AgentNetTool records screen video, mouse, keyboard and a11y tree on personal PCs.
- **Signal:** supervised: human action per step plus synthesised L1 / L2 / L3 reasoning.



Our redraw (not the paper's figure): the demo-to-policy pipeline and the reasoning levels.

$$P(a_i \mid I, s_0, a_0, \ldots, s_i)$$

Policy over the next action given instruction and history (Sec. 2.1)

Key results: OSWorld-Verified success rate (%), online

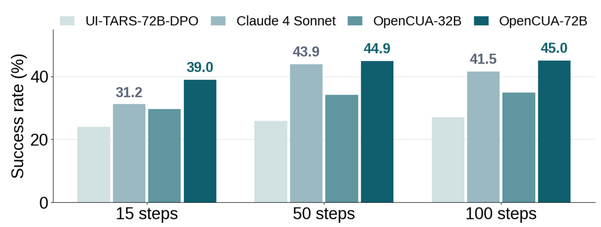


Table 3, redrawn; competitors as printed there.

72B: 39.0 / 44.9 / 45.0 vs Claude 4 Sonnet: 31.2 / 43.9 / 41.5

SO WHAT → *Claude Sonnet 4.5: 61.4% at 100 steps (Table 3, not charted).*

Ablation: OSWorld success rate (%), Qwen2-VL-7B

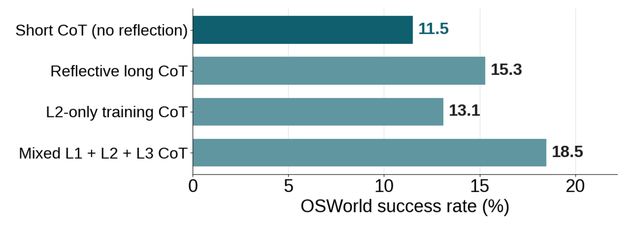


Table 6: CoT variants; the two pairs come from separate runs and data sizes, so compare within a pair.

SO WHAT → *Reflective long CoT lifts 11.5 to 15.3 (Qwen2-VL-7B, 14K Win&Mac + 3K Ubuntu); mixing L1 / L2 / L3 beats L2-only (18.5 vs 13.1). The paper conjectures the gain comes from better self-correction. Test-time format: L2 18.5, L3 17.6, L1 16.9.*

Headline Numbers

45.0%
OSWorld-Verified, online
OpenCUA-72B, 100 steps

- **34.8%** OpenCUA-32B, 100 steps
- **22.6K** human demos
- **60.8%** ScreenSpot-Pro offline

The paper's own controls

CONTROL	OPENCUA	COMPARED WITH
Steps 15 / 50 / 100 (32B)	29.7 / 34.1 / 34.8	Sonnet 4: 31.2/43.9/41.5
Pass@3 vs Pass@1, 100 steps	7B 36.48, 32B 45.10	7B 26.67, 32B 34.88
Offline AgentNetBench avg	32B 79.1, 7B 75.2	OpenAI CUA 73.1
Data cost (USD)	about 32,000 total	CoT 0.6 per task

Offline is not online; Pass@3 is not Pass@1.

Takeaway

22.6K human demos plus reflective long CoT give open models 45.0% (72B) on OSWorld-Verified; ablations use Qwen2-VL-7B.

Sonnet 4.5: 61.4% at 100 steps

Ablations on Qwen2-VL-7B

Release announced

Scan to Read

Paper

Code